MODULE 2 · LECTURE 4

CS 6530 Applied Cryptography

Conventional Cryptography and Symmetric-Encryption Foundations

Module 2 – Symmetric Cryptography, Hashes and Authentication

COURSE ROADMAP

Where Module 2 Fits in the Course

1. **Module 1 –** Foundations of Applied Cryptography
2. **Module 2 – Symmetric Cryptography, Hashes and Authentication**
3. **Module 3 –** Public-Key Cryptography and ECC
4. **Module 4 –** Key Management, Certificates and PKI
5. **Module 5 –** IPsec, IKE and Secure Gateways
6. **Module 6 –** Post-Quantum Cryptography
7. **Module 7 –** TLS and Secure Administration
8. **Module 8 –** Secure Design and Deployment Applications

SECURITY BY APPLICATION TYPE

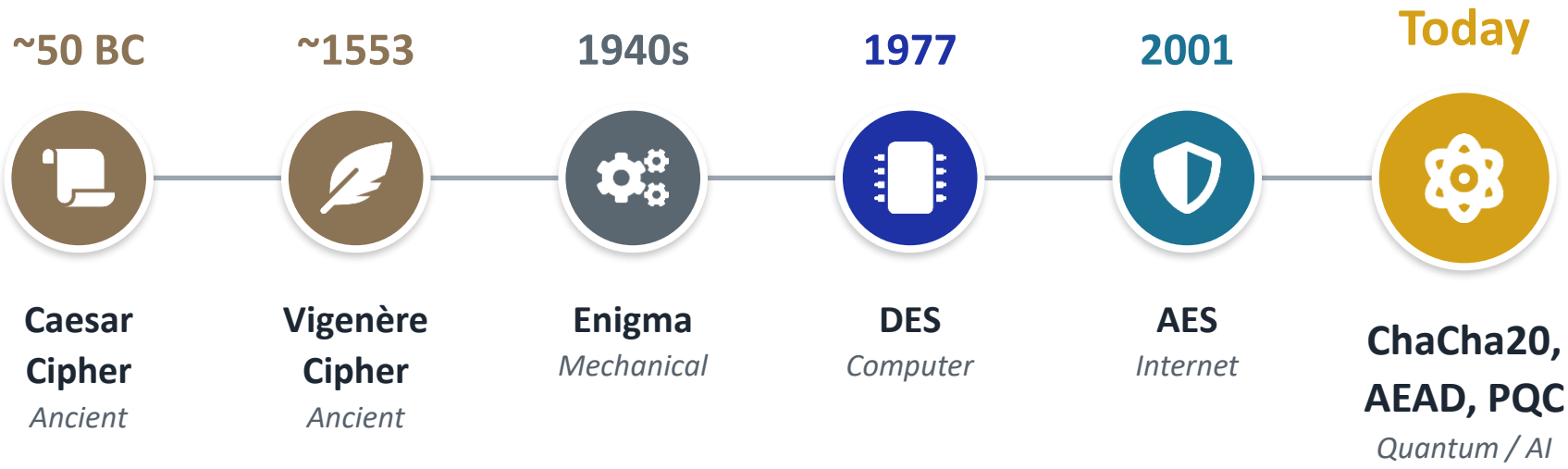
Modern Applications Have Different Security Requirements

Application Type	Example Systems	Nature of Data	Security & Engineering Reqs.	Why Security is Required
Secure Messaging	Signal, WhatsApp	Text, images, voice, video	**C I** *(Low latency)*	Prevent eavesdropping and message tampering
Web Applications	HTTPS	Web pages, APIs, downloads	**C I** *(Variable-size objects)*	Secure Internet communication
Network Security	IPsec VPN	Continuous network traffic	**C I** *(High throughput)*	Protect enterprise traffic over untrusted networks
Wireless Security	WPA2 / WPA3	Wireless frames	**C I A** *(Low overhead)*	Prevent unauthorized access and frame tampering
Mobile Networks	5G User Plane	Continuous packet flow	**C I** *(Low latency, ordered delivery)*	Secure real-time mobile communication
Storage Security	BitLocker, LUKS	Data at rest	**C** *(Random access)*	Protect stored data on lost/stolen devices
Cloud Security	AWS / Azure Storage	Data objects	**C I** *(Scalability)*	Protect cloud data and satisfy compliance requirements
Streaming & Media	Netflix DRM	Continuous media streams	**C** *(High throughput)*	Prevent unauthorized content access

Discussion: *Do all these applications require exactly the same cryptographic solution?*

CRYPTOGRAPHIC EVOLUTION

The Evolution of Cryptography: Responding to New Challenges



Cryptography continuously evolves in response to changing communication technologies, advances in computing, and increasingly capable attackers.

EVOLUTION OF CIPHER FAMILIES

The Evolution of Classical Cryptography

Substitution *(Replace Symbols)*
Replace one symbol with another using a secret rule.
Caesar, Monoalphabetic

Transposition *(Rearrange Symbols)*
Rearrange symbol positions while preserving the original symbols.
Rail Fence, Columnar

Polyalphabetic *(Vary Substitutions)*
Use multiple substitution alphabets controlled by a key.
Vigenère

Mechanical *(Mechanize Encryption)*
Automate encryption using rotating mechanical components.
Enigma

Perfect Secrecy *(Use True Randomness)*
Encrypt using a truly random one-time key.
Vernam / One-Time Pad

Each new cipher family introduced a different encryption idea to improve security and address the limitations of earlier approaches.

CLASSICAL CIPHER DEMONSTRATIONS

Representative Classical Ciphers – Demonstrations

Cipher Family	Representative Cipher	Core Encryption Idea	CrypTool Output
Substitution	**Caesar**	Replace each symbol using a fixed shift.	DSSOLHGFUBSWRVWXGHQWVDUHLQWHOOLJHQW **APPLIEDCRYPTOSTUDENTSAREINTELLIGENT** *Caesar Decryption (Shift = 3)*
Transposition	**Rail Fence**	Rearrange symbol positions (Rails = 3).	AIRODSILEPLECYTSUETAENELGNPDPTNRTIT
Polyalphabetic	**Vigenère**	Multiple substitution alphabets (Key = SMART).	SBPCBWPCIRHFOJMMPEEMKMRVBFFECEASEEM
Mechanical	**Enigma**	Rotor I-II-III, Reflector A, Plugboard AB CD.	JOGATWRJNCWXNNYZCPHSJENJPGRQAQVQNVU
Perfect Secrecy	**Vernam / OTP**	One-time key: THISCOURSEISTOMAKESECU REDESIGNENGER	TWXDKSXTJCXLHGFUNIFXUUIILRLMRYMTKRK

The same plaintext produces different ciphertext because each cipher family applies a different encryption principle.

ENGINEERING EVOLUTION

Engineering Evolution: From Classical to Modern Cryptography

1 · ENGINEERING EVOLUTION



Mainframe
Centralized Computing

Desktop
Personal Computing

Laptop
Mobility

Smartphone
Ubiquitous Connectivity

Message: *Technology evolves to meet changing requirements.*

2 · EVOLUTION OF CIPHER TECHNOLOGY



Caesar
Simple Substitution

Vigenère
Multiple Alphabets

Enigma
Machine-assisted Encryption

DES
Electronic Cryptography

AES
Modern Block Cipher

ChaCha20
Software-efficient Stream Cipher

Message: *Cryptography evolved as communication systems, computing capability and security requirements changed.*

Engineering evolves by learning from earlier designs and adapting to new requirements. Modern cryptography is one example of this enduring engineering principle.

ENDURING ENGINEERING LESSONS

What Cryptographic History Teaches Engineers

Kerckhoffs' Principle — A secure cryptographic system should remain secure even when its design and algorithms are publicly known. Only the cryptographic key needs to remain secret.

Publicly Reviewed Design
Security improves through public analysis, independent evaluation and continuous improvement.

Key-Centric Security
Protect the key, not the algorithm.

Adapting to New Requirements
New applications require new cryptographic designs.

Continuous Evolution
Security engineering evolves together with technology.

Open Design → **Extensive Security Analysis** → **Weaknesses Identified** → **Improved Cryptographic Algorithms**

Why This Matters: Public review strengthens cryptographic systems by enabling researchers worldwide to identify weaknesses, improve designs and build confidence before deployment. Modern standards such as AES, SHA-3 and emerging post-quantum algorithms are products of this open engineering process.

The Modern Symmetric-Key Model



Engineering Principle: The algorithm may be publicly known. Security depends on protecting the secret key.

Classical: Algorithm Secret	**Modern:** Key Secret

Modern symmetric cryptography relies on public algorithms and secret keys, enabling secure and efficient protection of large volumes of data.

MODERN SYMMETRIC CRYPTOGRAPHY

Why Symmetric Cryptography Dominates Modern Communication

WHY SYMMETRIC?

Fast
High-speed encryption/decryption

Efficient
Low computational overhead

Scalable
Suitable for large volumes of data

Practical
Enables real-time secure communication

REPRESENTATIVE APPLICATIONS

Messaging	*Messages & Media*
HTTPS	*Web Data*
5G User Plane	*User Traffic*
VPN	*Tunnel Traffic*
Streaming	*Multimedia Streams*
Cloud Storage	*Stored Data*

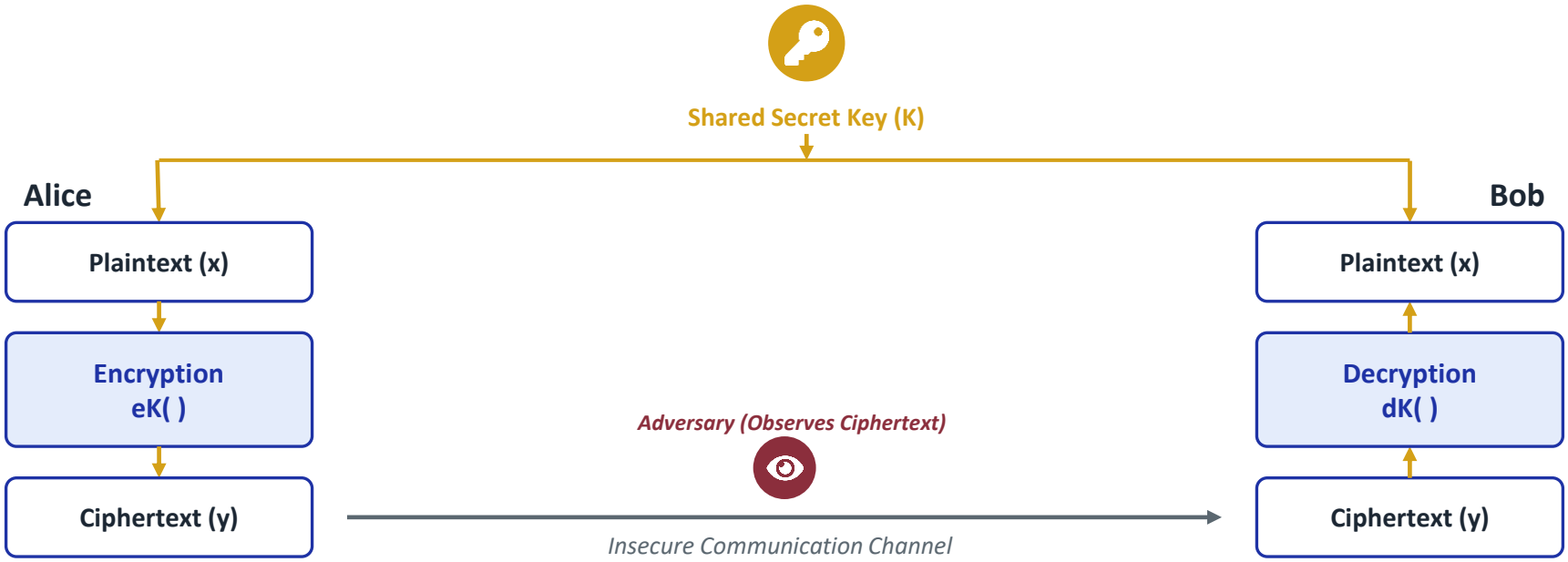
ENGINEERING REALITY



After a secure key has been established, the actual application data is protected using symmetric cryptography.

Symmetric cryptography is the workhorse of modern communication because it provides fast, efficient and scalable protection for application data.

The Symmetric Cryptographic Communication Model



`x`	Plaintext	`y`	Ciphertext	`K`	Shared Key	`eK(.)`	Encryption	`dK(.)`	Decryption

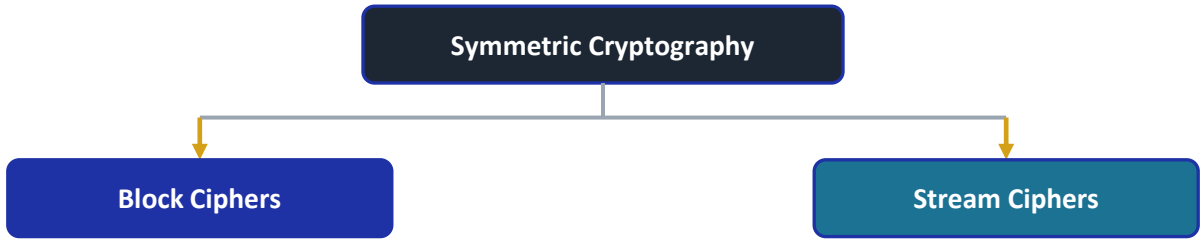
$$y = eK(x) \qquad x = dK(y)$$

Engineering Observation: Both parties must possess the same shared secret key. Without it, recovering the plaintext should be computationally infeasible.

Symmetric cryptography enables secure communication over insecure channels using a shared secret key.

MODERN SYMMETRIC CRYPTOGRAPHY

The Two Major Families of Symmetric Algorithms



Block Cipher Family

Processes fixed-size blocks of data.

DES · 3DES · AES

Typical Applications:

File Protection · Storage Encryption · Secure Records

Stream Cipher Family

Processes data as a continuous stream.

RC4 · Salsa20 · ChaCha20

Typical Applications:

Voice · Video · Streaming · Interactive

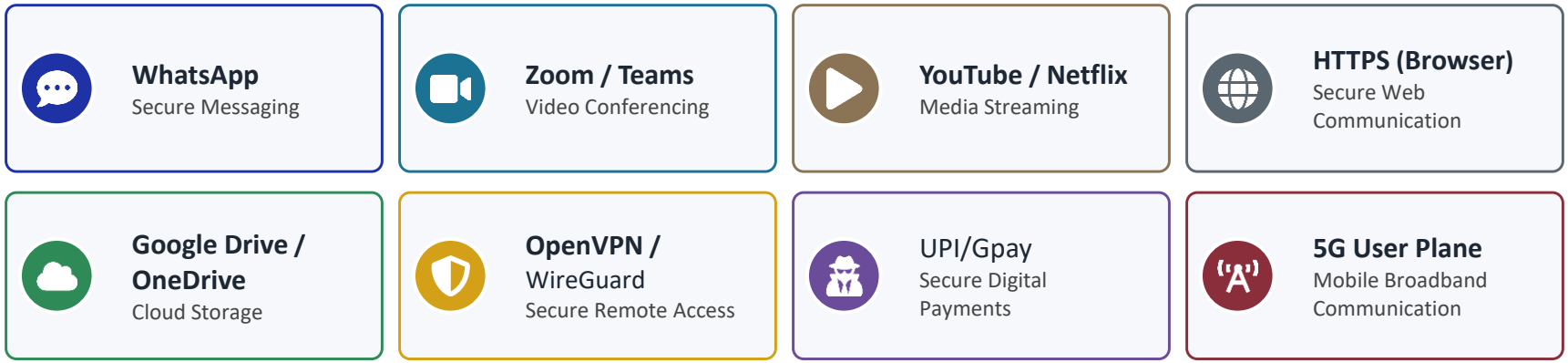
Engineering Aspect	Block Cipher	Stream Cipher
Data Processing	Fixed-size blocks	Continuous data stream
Representative Algorithms	DES, 3DES, AES	RC4, Salsa20, ChaCha20
Typical Applications	Files, Storage, Secure Records	Voice, Video, Streaming

Engineering Observation: Both families provide symmetric encryption. The choice depends on application requirements and data characteristics.

Block ciphers and stream ciphers are complementary families of modern symmetric cryptography, chosen by application and data characteristics.

ENGINEERING DESIGN THINKING

Think Like a Security Engineer



Imagine you are the security engineer responsible for one of these applications. What should be your next engineering step?

ENGINEERING DECISION WORKFLOW



Understand the application → Identify the requirements → Analyse the threat model → Select the appropriate cryptographic solution.

Applied Cryptography begins with understanding the problem. Cryptographic algorithms are engineering tools selected to satisfy application requirements.

LECTURE WRAP-UP

Summary and Looking Ahead

TODAY WE LEARNED

1. Why cryptography evolved
2. Evolution of classical cipher families
3. Kerckhoffs' Principle
4. Modern symmetric cryptography
5. Symmetric communication model
6. Block and Stream cipher families
7. Engineering design thinking

LOOKING AHEAD — NEXT LECTURE

Feistel Networks | **DES** | **3DES** | **AES**

Assignment 1: Secure Data Protection using AES-GCM and ChaCha20-Poly1305

Details will be available through Moodle / course announcements.

Understanding the problem comes first. Engineering solutions follow. Modern cryptography is one of those engineering solutions.